Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

Some **Key areas protected by private FP&S service providers or businesses**

FP&S: Fire prevention and safety

Targeted installation and commissioning done much earlier for these customers or departments

Solo use of FP&S equipment or systems is on a purchase, training and/or AMC basis

These service providers serve incidences and requirement more than impactful needs for RS&A - RCS

These service providers are dependent on case scenario relative response

These service providers use products based approaches

But can instantiate RSA-RCS packaging for FES

For this issue detection is RSA-RCS EOCC linked priority or intelligence based as Prepared FP&A systems may not be available

Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

<u>Critical locations that are ;protected by governmental departments of Fire & Emergency Services</u>

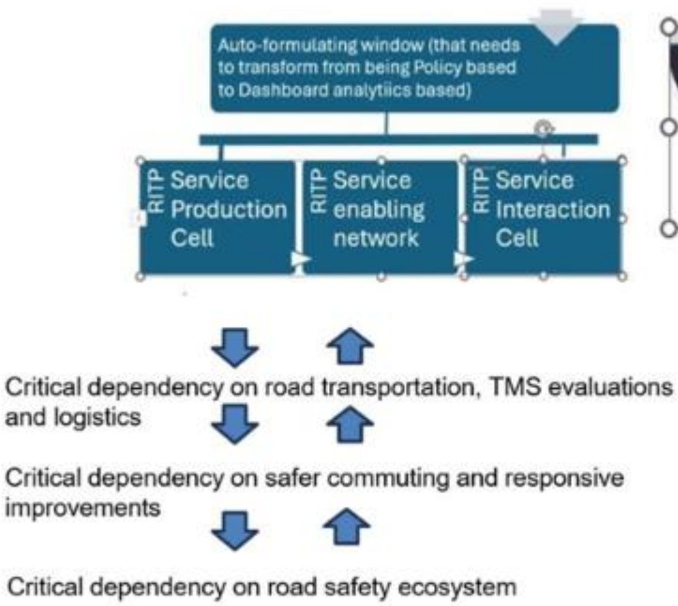
- ❑ Roads & Road Systems
- ❑ Road Arboriculture
- ❑ Road Infrastructure
- ❑ Traffic Engineering (TE)
- ❑ Traffic Control deployments
- ❑ Immersive TE deployments
- ❑ Commuter safety deployments
- ❑ Goods transportation networks/supply chains
- ❑ BSNL/TELECOM/similar deployments
- ❑ BESCOM /similar deployments
- ❑ BWSSB / similar deployments
- ❑ Healthcare services deployments
- ❑ Medical supply services
- ❑ Educational institutions/in-situ sites/campuses
- ❑ EV Infrastructure/Flexi-fuel pumps
- ❑ Automobile dealer networks
- ❑ Automobile service centres/businesses
- ❑ Mobile Vehicle Assistance Units (MVAU)
- ❑ MSME Manufacturers
- ❑ MSME Service deployments
- ❑ Corporate offices/campuses
- ❑ KSFES deployments
- ❑ FESA auditable buildings/sites/complexes
- ❑ Fast Track PRM deployments

These FES service providers are dependent on Responsive Commuting Services all the time

These service providers use packaging of FES response and need to design right approaches

These service providers are called upon or expected to respond to instantiate RSA-RCS packaging for FES

For this, issue detection needs to be developed through a policy called Unsecured Responsiveness



Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

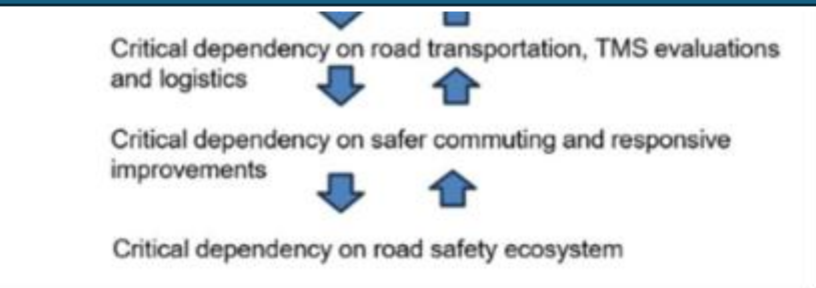
For both unsecured responsiveness and RSA-RCS EOCC linked responsiveness, the implementation of RS&A V2R effectiveness needs to be developed in specific interests such as

Afflicted / Dependent Risk Mitigation Intelligence staged across Production – Instantiation – Trained or Tested Responsiveness – Advanced Alpha Assistance – Call to attention Services & Assistance

To do this, the RSA-RCS teams will need to package the responsive services and assistance using case scenario based

- Target bands for V2R effectiveness
- Decisions to opt for REGULAR RS&A-RCS or SMART RS&A-RCS packages
- Sources of information
- Methodology Hypothesis via earlier Hypothesis Testing using case studies, empirical studies or surveys
- SMART Methodology Tabulations
- SMART Methodology Training data
- SMART Methodology Testing data
- SMART Methodology Deprecated data
- SMART Methodology Future Forward Contracting or EOCC linked RS&A-RCS

Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization



<u>Key considerations of commercial vehicles</u>

- **Gross Vehicle Weight (GVW):** GVW determines the vehicle's capacity for carrying goods or passengers.
- **Fuel Type:** Commercial vehicles can be powered by diesel, petrol, CNG, or electric.
- **Payload Capacity:** This is the maximum weight a commercial vehicle can carry, for simpler understanding, when the vehicle in used for transporting goods.
- **Wheel base, Number of wheels/tires (known to use** radial or bias-ply construction)
- **Ground clearance: t**he distance between the ground and the lowest point of the vehicle, measured in millimeters.
- **Steering type: Hydraulic power assisted**
- **Safety systems: ABS, ESC, Designed for driver comfort, Build quality to mitigate crash impact**
- **Key specifications** include engine type, capacity, power output, torque, fuel tank capacity, transmission, brakes, suspension, tire size, wheelbase, and overall dimensions (length, width, height).
- **Awareness and Responsiveness to** ensure safe, well-maintained and optimally performing vehicle, assist and safety systems

Sensitized and Effective decision making

Driving condition responsiveness refers to how quickly and appropriately a driver reacts to changing road and traffic conditions. This includes their reaction time to hazards, their ability to maintain control of the vehicle, and their decision-making processes in varying scenarios. Factors like stress, fatigue, and driver ability can significantly impact this responsiveness.

Key Aspects of Responsiveness

Enforcement Directive networked
Tachograph for Driving Time, Working Time and Rest periods

Reaction Time
Control over the vehicle
Defensive Driving practices
Sensitized or Effective Decision-making

Key Factors affecting Responsiveness

Driver Training/Evaluations
Driver Ability/Anticipatory Guidance
Driver Stress
Driver Alertness / Fatigue
Road and Traffic conditions
NSSR-RS Training Programme, Pre & Post ride checklists, instructions
Enforcement Directives for DT, WT and Rest
DT, WT and Rest Timings surveyed per state and across state borders

Defensive Driving (Areas of importance for safer and sustainable commuting by commercial vehicles)

- ❑ Safe and Fuel-efficient driving
- ❑ Awareness, knowledge and understanding of technical characteristics, benefits and operations of safety controls
- ❑ Limits and conditions for the use of brakes
- ❑ Better use of speed and gears
- ❑ Ways of slowing down and braking on downhill stretches, and factored decisions or actions to be taken in the event of a mechanical or active safety system failure
- ❑ Ability to help passengers find possible ways for safety and effectively use public & private infrastructure
- ❑ Dedicated Lane driving
- ❑ Managing conflicts between safe driving and other roles as a driver
- ❑ Interactions with passengers
- ❑ Knowing and understanding specific ness of certain passengers/ and target groups like children, the disabled, the commuters needing alpha assistance
- ❑ Knowledge and maximum working periods (Driving time, working time and rest) with the use of a tachograph

Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

Defensive Driving (Areas of importance for safer and sustainable commuting by commercial vehicles)

- ❑ Knowledge of passenger expectations, luggage, need for safety equipment, need for seat belts, lap belts and other anchorage
- ❑ Knowledge of vehicle load rules and causation factors
- ❑ Awareness of traffic signs, road traffic risks, types of accidents in the transport sector and driving condition responsiveness, accident statistics, accident causation factors, consequences to vehicle, functional expectations, equipment, organizational and public human resources, materials, money involved
- ❑ Ability to respond to emergencies, concerning situations, accident-causing factors via training, and responsiveness for procedures of First Aid, Fire Evacuation/ Fire Detection & Response potential driver / passenger conflict situations, implications to the safety of passengers/commuters and the impactful issues for road safety
- ❑ Understanding of the importance of physical and mental wellbeing for road safety,, with the opting for "of a healthy lifestyle/diet, and avoidance of alcohol, drugs and other substances, and interests for decisions" known to affect behaviour/alertness, the adherence to work-rest cycles known to cause fatigued conditions

ii. **Active Safety Systems that universally help**

- ❑ Electronic Stability Control (ESC)
- ❑ Tyre Pressure Monitoring System
- ❑ Cruise Control / Adaptive Cruise Control
- ❑ Advanced Braking System / ABS
- ❑ Collision Avoidance Warning Systems / Forward Collision Warning Systems
- ❑ Lane Departure Warning Systems
- ❑ Driver Alert Control Systems
- ❑ Reverse Assist Systems
- ❑ Fuel Monitoring System

- ❑ Charter for Driver Fitness/Driver Training, Vehicle Fitness and Performance Analysis, Road System understanding, Alpha Assistance
- ❑ Periodic Inspection
- ❑ Pre-ride and Pot-ride checklists for fitness and system analysis
- ❑ Corrective Maintenance

Responsiveness for A-S-P and DI in operational environment
A: Awareness
S: Sensitization
P: Preparedness
DI: Deep Interaction

Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

Types of buses

- **City Buses**
- **Inter-city Buses**
- **Suburban Buses**
- **Tour Buses**
- **Staff & Contract Buses**
- **School Buses**

Key considerations for buses

- **Primary Function:** Passenger transport.
- **Capacity:** High passenger capacity, often with multiple rows of seats.
- **Features:** Comfort-focused seating, often with amenities like air conditioning and entertainment systems.
- **Value addition:** Fuel Monitoring System, Cruise Control, Gear Shift Advisor, Firmware over the air, Telemetry
- **Safety features:** Blind Spot Mirror, Anti-skid flooring, Panic button, Guard Rails, Emergency brakes with ABS

Value addition functions in buses

- ❑ Fuel Monitoring System
- ❑ Improved Engine Hood Installation
- ❑ Pagoda Gear Box
- ❑ Gear Shift Advisor
- ❑ E-Viscous Fan for engine cooling
- ❑ Cruise Control
- ❑ Gateway Domain Controller Unit for seamless communication, data exchange, and security
- ❑ Firmware over the air
- ❑ Fleet Management Telemetry
- ❑ USB Charging Ports

Need for Intelligent Transport Systems, Navigation, Perspective Vision and Feedback Systems that improve road system understanding for concerns like height and weight restrictions for vehicles, planning for pretexts such as bridges, tunnels, ring roads, flyovers, under-passes, need for emergency call features on route

Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

Safety system functions in buses

❑ Top Maker Lamp

❑ Blind Spot Mirror, Indirect Vision Detection

❑ Heavy Duty Wiper

❑ Emergency Exit, Service Doors, Peep Windows

❑ Transparent Door

❑ Guard Rails

❑ Anti-skid flooring

❑ Panic button

❑ Safe Seat Upholstery with Grip Handle

❑ Seat Belt Anchorage, Lap Belt, Impact Shield Instead of Air bags

❑ Hand hold Straps

❑ LED Destination Board

Safety system functions in buses

[ ] Fog Lamp
[ ] Reverse Parking Sensor
[ ] 7 Layer Exhaust Insulation
[ ] Fire Detection and Alarm System Indicators (in Engine Compartment, Passenger Areas and Luggage Areas)
[ ] Fire Detection Systems with Cable test for satisfactory behaviour, Repel test for insulation, Burning material behaviour tests
[ ] All Wheel Disc Brakes
[ ] Emergency Braking with ABS
[ ] Gear Level with Cable Shaft
[ ] Speed Deceleration Stop Lights
[ ] DRL Head/Tail Lamps
[ ] Unitized wheel bearing
[ ] Robust Multi-piece Propeller Shafts

Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

Case scenario-based inclusions in the RS&A-RCS packaging

Ticked as appropriate

- For the RS&A Network or departments, the **decision making for V2R effectiveness can need to identify, assess and address**
- **1.** RS&A RCS **Vehicle Parts/Components/Systems/Panels classification and codification**
- **2.** RS&A RCS V2R **Process planning, Process variation**
- **3. Discrete Events and RS&A PI factors**
- **4. Human factors**
- **5. Machine utilization factors**
- **6. RS&A PI control/Process parameters**
- **7. Movement / Motion study**
- **8. SMART RS&A Packaging / V2R codification (basic functions)**
- **9. Working range of RS&A Packaged workflows/lifecycles/processes/functions**
- **10. Guidelines for RS&A PI conformity**
- **11. Factors adding to RS&A Package Setup time**
- **12. RS&A Packaged Workflow or V2R Lifecycle Sequence/EOCC Sequence**
- **13. RS&A PI efficiency factors or V2R effectiveness**
- **14. NADL standardized** ***Diagnostics to control variance in responsiveness - /improve RS&A packaging quality***

Case scenario-based deviations are mitigated in the RS&A packaging

Ticked as appropriate

- MTTD: Mean Time to Detection
- MTTP: Mean Time to Prioritize
- MTTN: Mean Time to Network needed Engineering infrastructure and resources
- MTTR: Mean Time to Resolution
- MTTCOPQ: Mean Time to Cost of Poor Quality
- MTAAR: Mean Time to Alpha Assistance Resolution (for afflicted or aged commuters)

Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

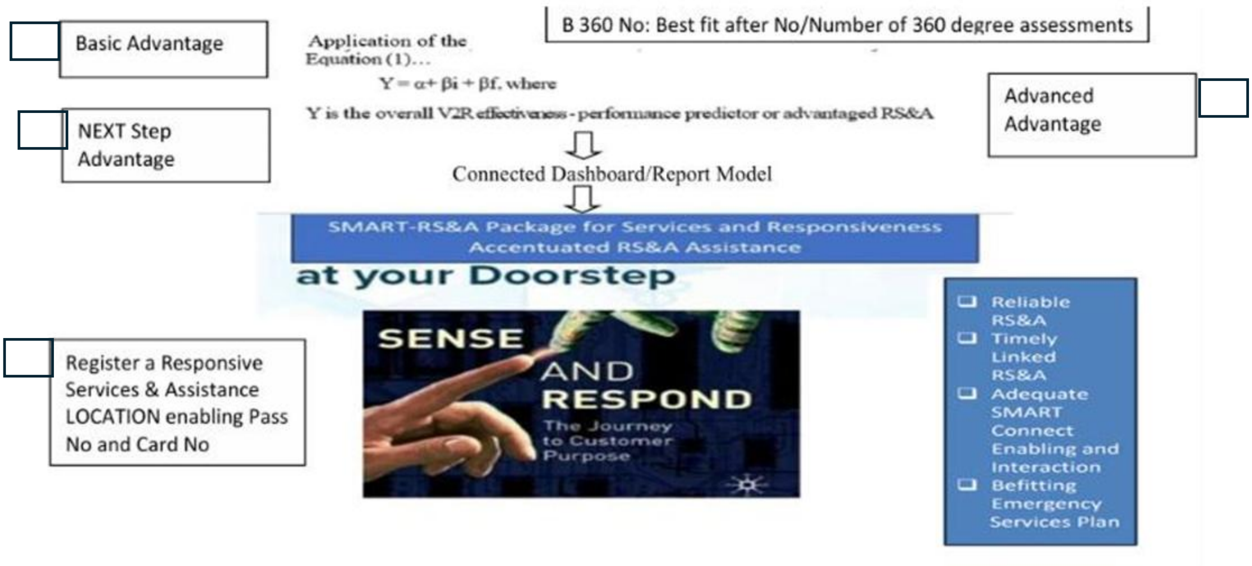
Case scenarios-based issues for road system understanding are included in the RS&A-RCS packaging

Ticked as appropriate

- 1. Unplanned Lanes, Road Medians
- 2. Unplanned Bordering Road Barricades
- 3. Unplanned Speed breakers or Road Humps
- 4. Unplanned or poorly constructed Pavements
- 5. Poorly maintained Manholes & Sewer systems
- 6. Impediment causing Elevated or Tube Railway infrastructure
- 7. Unmanned or poorly maintained Railway crossings
- 8. Poorly maintained Bridges and Tunnels*
- 9. Poorly maintained Trees and Greenery
- 10. Hotspots (locations that need converged administration to address the need to mitigate climate change, rising pollution levels, rising CO_2 levels, poor air quality, accident trends, traffic problems, incidences of crime, issues with road system arboriculture)
- 11. No Road Infrastructure Transformation evaluations to minimize RADIUS OF COVERAGE inefficiencies

Dependent elements used to design/prepare RS&A-RCS packaging, where these elements help V2R effectiveness, responsiveness and agility

Tick as appropriate



Bulletin Board to describe planning for RS&A-RCS packaging with vehicle utilization

Dashboard/report model elements used to design/prepare RS&A-RCS packaging, where these elements help V2R effectiveness, responsiveness and agility

Tick as appropriate

- (1) Package category: Descriptive name for the Package Code
- (2) Package Code: Package-provider-code or Package-Provider-code.Package-EOCCProvider-code
- (3) PrevV2R: V2R Effectiveness for the previous experiences
- (4) PrevV2RMin: Minimum Target Band for V2R Effectiveness for the previous experience
- (5) PrevV2RMax: Maximum Target Band for V2R Effectiveness for the previous experience
- (6) PrevV2RAmbiguity: Whether this Packaging for V2R effectiveness was clear or ambiguous
- (7) PrevV2RConsensus: Whether this Packaging for V2R effectiveness had a consensual agreement
- (8) PrevV2RAvailability: Whether this Packaging for V2R effectiveness was consistently available

- (9) CurrentV2R: V2R Effectiveness for the current year
- (10) CurrentV2RMin: Minimum Target Band for V2R Effectiveness for thecurrentyear
- (11) CurrentV2RrMax: Maximum Target Band for V2R Effectiveness for thecurrentyear
- (12) CurrentV2RAmbiguity: Whether this Packaging for V2R effectiveness is clear or ambiguous
- (13) CurrentV2RConsensus: Whether this Packaging for V2R effectiveness has a consensual agreement
- (14) Currentv2rAvailability: Whether this Packaging for V2R effectiveness is consistently available
- (15) FilterV2RTopic: Whether this Packaging for V2R effectiveness is mandatory or filtered for topic analytics
- (16) FilterV2RTopic-Subtopic: Whether this Packaging for V2R effectiveness is mandatory or filtered for topic-subtopic analytics
- (17) Dept-responsible-for-V2RTopic: The name of the department/team responsible for Packaging for V2R effectiveness for topic analytics
- (18) Dept-responsible-for-V2RTopic-Subtopic: The name of the department/team responsible for Packaging for V2R effectiveness for topic-subtopic analytics